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Spatiotemporal Traffic Congestion Patterns and Network Centrality in Indonesian Metropolitan Cities

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Abstract

Urban traffic congestion poses significant challenges to the sustainable development of~~in~~ rapidly growing cities in~~across~~ Southeast Asia. This paper~~study~~ presents a comprehensive spatiotemporal analysis of traffic congestion patterns in three major Indonesian metropolitan areas: Jakarta, Bandung, and Semarang. Utilizing high-resolution traffic flow data collected from the HERE Traffic API over an 11 -month period (March 2025-~~to~~ February 2026), we analyzed over 264 million traffic observations across 18,694 road segments. The methodology integrates real-time traffic jam factor measurements with geostatistical analysis (Moran's I, LISA, Getis-Ord Gi*), OpenStreetMap network analysis using OSMnx, and a novel graph-based capacity-drop detection framework. Our most significant finding is that temporal factors dominate spatial factors by a factor of 1,000−4,000 × in explaining congestion variance;~~:~~ time-of-day ($\eta^2 = 15-24\%$) vastly outweighs network centrality ($R^2 < 0.01\%$) and POI density ($R^2 < 0.01\%$) as predictors. A dedicated bottleneck analysis further confirms this conclusion: road capacity scores show negligible correlation with congestion ($r < 0.04$), spatial capacity drops (where road hierarchy decreases) show no proximity effect on congestion ($p = 0.10-0.83$,

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|d| < 0.09), and local bottleneck gradients are indistinguishable from the surrounding segments (|d| < 0.05). This demonstrates that the congestion is ~~fundamentally~~ a temporal synchronization problem ~~-~~resulting from millions of people traveling simultaneously, rather~~at the same times rather~~ than a spatial infrastructure or capacity constraint problem. The evening peak period (16:00−19:00) shows congestion approximately 40% higher than the daily average~~s~~ across all cities. While global spatial autocorrelation is not non-significant, LISA identifies local hotspots representing peak-hour capacity bottlenecks. These findings support the prioritization~~prioritizing~~ of demand management strategies (staggered hours and~~,~~ flexible work) over infrastructure expansion, contributing new empirical evidence to traffic policy debates in rapidly urbanizing contexts.

Word count: 6,261 words, excluding references.

Keywords: Urban traffic congestion, Spatiotemporal analysis, Network centrality, Jam factor, Capacity bottleneck, Indonesian cities, HERE Traffic API, OSMnx, Spatial autocorrelation

1. Introduction

1.1. Background

Urban traffic congestion has emerged as one of the most pressing challenges facing the rapid~~ly~~ urbanization~~urbanizing~~ of cities in developing countries [28]. In Southeast Asia, where urbanization rates exceed global averages, traffic congestion imposes substantial economic costs, estimated at 2−5% of GDP annually [4]. Indonesia, ~~as~~ the world's fourth most populous nation with over 270 million inhabitants, exemplifies these challenges, with its major metropolitan areas experiencing severe and worsening congestion conditions.

The three cities examined in this study, ~~-~~Jakarta, Bandung, and Semarang, represent distinct urban typologies within the Indonesian context. Jakarta, the national capital with a metropolitan population exceeding 34 million, consistently ranks among the world's most congested cities [35]. Bandung, with approximately 2.5 million residents, serves as Java's second-largest city and ~~a~~ major educational and industrial center. Semarang, home to 1.8 million people, functions as Central Java's capital and a key logistics hub connecting Java's northern coast.

1.2. Research Gap and Objectives

Although~~While~~ previous studies have examined traffic congestion in Indonesian cities using aggregate or survey-based approaches [33, 19], limited research has employed high-resolution, real-time traffic data for systematic spatiotemporal analysis. Furthermore, the integration of network

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topology metrics with traffic flow data remains underexplored in ~~the~~ Indonesia~~n context~~.

This study addresses these gaps by:

1. Analysis of~~Analyzing~~ traffic congestion patterns using continuous high-frequency traffic flow data over an extended temporal period
2. Applying geostatistical methods to identify spatial clustering and hotspot patterns
3. Integrating OpenStreetMap network analysis to examine relationships between network centrality and congestion distribution
4. Providing comparative insights across cities of varying scales and urban morphologies

1.3. Contributions

This research makes the following contributions to the literature:

- Methodological: This study d~~D~~emonstrates the integration of commercial traffic API data with open-source network analysis tools for comprehensive urban traffic assessment, including a novel graph-based capacity drop detection framework.
- Empirical: This is~~Provides~~ the first systematic comparative analysis of traffic patterns across three major Indonesian cities using standardized metrics, with triangulated evidence from four independent spatial predictors (centrality, POI density, road capacity, and capacity drops) ~~all~~ yielding null results.
- Theoretical: Demonstrates that normalized congestion metrics (jam factor) inherently remove capacity effects, with implications for capacity congestion studies using similar data sources.
- Practical: Offers evidence-based recommendations for traffic management and infrastructure prioritization in rapidly urbanizing Southeast Asian contexts.

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2. Literature Review

2.1. Urban Traffic Congestion in Developing Countries

Traffic congestion in developing countries exhibits distinct characteristics compared with~~to~~ developed nations, including higher variability, more pronounced peak periods, and greater sensitivity to informal transport modes [13]. Studies in Asian megacities have documented congestion levels that significantly impact economic productivity and quality of life [10, 23]. The emergence of big data approaches has opened new possibilities for

understanding urban dynamics with~~at~~ unprecedented spatial and temporal granularity [6].

In the Indonesian context, traffic congestion has been extensively studied from both behavioral and policy perspectives. Susilo et al. [33] examined commuting patterns in Jakarta and~~,~~ fou~~i~~nd~~ing~~ that the average commute times exceeded 90 min~~utes~~ for many workers. Joewono and Kubota [19] analyzed public transport satisfaction and~~,~~ reveal~~ed~~~~ing~~ significant dissatisfaction related to congestion-induced delays. However, these studies relied primarily on survey data rather than continuous traffic measurements.

2.2. Traffic Flow Measurement and Analysis

The advent of probe vehicle data and commercial traffic APIs has transformed traffic analysis capabilities [21, 17]. The jam factor, a normalized congestion metric ranging from 0 (free flow) to 10 (complete standstill), has been~~become~~ widely adopted for cross-network comparison~~s~~ [16]. Studies utilizing similar metrics have successfully characterized congestion patterns in European cities [30] and examined spatiotemporal speed patterns in urban networks [12]. Recent studies~~work~~ have~~s~~ demonstrated that traffic jams propagate through urban networks in patterns analogous to simple contagion processes [32].

2.3. Network Analysis and Traffic

The relationship between network topology and traffic distribution has received increasing attention following the~~Boeing~~ [7] introduction of the OSMnx for street network analysis. Subsequent studies~~work has~~ extended network analyse~~i~~s to global urban contexts [9, 5]. Research has demonstrated correlations between centrality metrics, ~~-~~particularly betweenness centrality ~~-~~and traffic volume~~s~~ [14, 29]. Kirkley et al. [20] showed that network structure significantly influences congestion propagation, whereas~~while~~ topological analysis of urban street networks has revealed fundamental relationships between network form and function 18,24~~-~~, 25].

2.4. Traffic Bottlenecks and Capacity Constraints

The concept of traffic bottleneck ~~s~~-locations where road capacity decreases, causing flow breakdown and upstream queuing, ~~-~~ is central to the traffic engineering theory [11, 22]. Classical bottleneck models predict that capacity transitions (e.g., lane reductions and~~,~~ highway-to-arterial merges) concentrate on congestion at specific spatial locations [3]. However, empirical evidence linking static network capacity features to the observed congestion patterns in real urban networks remains limited. Most bottleneck studies focus on isolated highway segments rather than city-wide networks, and the interaction between normalized congestion metrics and capacity effects has received little attention.

2.5. Geostatistical Approaches to Traffic Analysis

Spatial autocorrelation methods, including Moran's I and local indicators of spatial association (LISA), have proven valuable for identifying traffic hotspots [27, 15]. The fundamental principle underlying these methods, -that near things are more related than distant things [34]-is particularly applicable to traffic networks where congestion propagates spatially. Studies have applied these techniques to examine crash patterns [1], urban spatial structure [37], and congestion clustering [36]. The PySAL library [31] has become a standard tool for implementing these methods in Python-based spatial analysis workflows.

3. Study Area and Data

3.1. Study Area Characteristics

The three study cities represent a hierarchy of Indonesian urban centers (Table 1).

Table 1: Study area characteristics

City	Population (million)	Area (km^2)	Traffic Segments	Urban Typology
Jakarta	10.5 (metro: 34)	662	14,549	Megacity, National Capital
Bandung	2.5	167	3,069	Large City, Regional Center
Semarang	1.8	373	1,076	Medium City, Provincial Capital

3.2. Data Collection

Traffic flow data were collected via the HERE Traffic API using bounding box queries at 30 -minute intervals from March 1, 2025, to February 6, 2026, yielding approximately 14,100 collection cycles per city (Table 2).

Table 2: Data collection summary

City	Collection Period	Total Files	Total Records	Unique Segments
Jakarta	Mar 2025 - Feb 2026	14,132	206,316,468	14,549
Bandung	Mar 2025 - Feb 2026	14,136	43,407,684	3,069
Semarang	Mar 2025 - Feb 2026	14,122	15,212,072	1,076
Total		42,390	264,936,224	18,694

3.3. Data Processing and Aggregation

Raw traffic data were aggregated into eight temporal periods to reflect~~ing~~ Indonesian urban activity patterns (Table 3).

Table 3: Temporal period definitions

Period	Time Range	Rationale
Night	00:00-05:59	Minimal activity
Morning Peak	06:00-08:59	Commute to work/school
Morning Off-Peak	09:00-11:59	Business hours
Lunch Hours	12:00-13:59	Midday break period
Afternoon Off-Peak	14:00-16:59	Business hours
Evening Peak	16:00-18:59	Return commute
Evening Off-Peak	20:00-21:59	Evening activities
Late Night	22:00-23:59	Reduced activity

lection, filtered for driveable roads (Table 4).

3.4. Network Data

Street~~-~~network data were obtained from OpenStreetMap using OSMnx [7]. Networks were downloaded using the same bounding boxes as in the traffic data collection and~~,~~ filtered for driveable roads (Table 4).

3.5. Data Quality Validation

Comprehensive exploratory data analysis confirmed the data integrity across all 24 dataset combinations (three~~3~~ cities × 8 time periods). Zero null

values were found for~~in~~ the key variables (jam factor, geometry, and observation count). All jam factor means fell within the expected 0−10 range, with per-segment means ranging from 0.00 to 2.29 across all cities. The s~~S~~egment counts are internally consistent across all

Table 4: Network statistics				
City	Nodes	Edges	Street Density (km/km^2)	Mean Degree
Jakarta	~ 45,000	~ 95,000	18.4	2.89
Bandung	~18,000	~ 38,000	22.1	2.76
Semarang	~12,000	~ 25,000	15.2	2.81

temporal periods within each city, and the observation density wa~~i~~s remarkably uniform (~ 1,766 observations per segment), confirming consistent data collection
methodology regardless of city size. Detailed validation tables are provided in the Supplementary Material S1.

4. Methodology

4.1. Analytical Framework

Our analytical framework integrates four complementary approaches (Figure 1):

1. Descriptive statistical analysis of temporal congestion patterns
2. Geostatistical analysis of spatial congestion clustering
3. Network analysis examining topology-congestion relationships
4. Bottleneck analysis testing capacity constraint effects on congestion

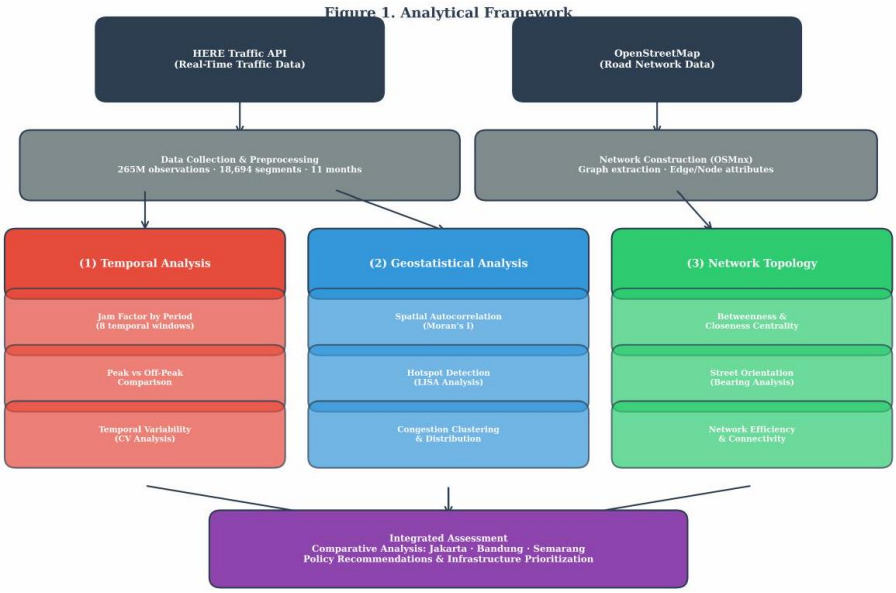

Figure 1: Analytical framework integrating temporal analysis, geostatistical methods, network topology assessment, and bottleneck capacity analysis.

4.2. Temporal Pattern Analysis

Temporal patterns were analyzed by computing the summary statistics for each defined period across all road segments. Analysis of variance (ANOVA) was em~~-~~ployed to test for significant differences between periods, with post-hoc Tukey HSD tests identifying specific ~~period~~ pairs of periods with significant differences. Diurnal patterns were visualized using hourly aggregations, while weekly patterns were used to examine~~d~~ day-of-week variations.

4.3. Geostatistical Analysis

4.3.1. Global Spatial Autocorrelation

Spatial autocorrelation was assessed using Moran's I statistic [26]:

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$$

where n is the number of spatial units, w_{ij} is the spatial weight between units i and j, and x_i is the jam factor of~~at~~ unit i.

4.3.2. Local Indicators of Spatial Association

Local Moran's I [2] identified specific hotspot and coldspot clusters:

$$I_i = \frac{(x_i - \bar{x})}{\sigma^2} \sum_j w_{ij}\left(x_j - \bar{x}\right)$$

4.3.3. Coefficient of Variation Analysis

Temporal stability was assessed using the coefficient of variation (CV):

$$CV = \frac{\sigma}{\mu} \times 100\%$$

Low CV values indicate consistent congestion levels, whereas~~while~~ high CV values suggest~~s~~ variable conditions.
4.4. Network Centrality Analysis

Following Boeing [7], we computed network centrality metrics:
 b~~B~~etweenness centrality measures the fraction of shortest paths passing through each edge.~~:~~

$$C_B(e) = \sum_{s \neq t} \frac{\sigma_{st}(e)}{\sigma_{st}}$$

where σ_{st} is the total number of shortest paths from node s to node t, and σ_{st} (e) is the number passing through edge e.

Closeness centrality measures average distance to all other nodes:

$$C_C(v) = \frac{n-1}{\sum_{u \neq v} d(u,v)}$$

where d(u,v) is the shortest path distance between nodes u and v.
Correlation analysis was used to examine~~d~~ the relationships between centrality metrics and observed congestion levels.

4.5. Bottleneck and Capacity Drop Analysis

To test whether physical road capacity constraints explain congestion patterns, we developed a multi-scale bottleneck analysis framework combining aggregate capacity scoring, graph-based capacity drop detection, and local gradient analysis.

4.5.1. Road Capacity Scoring

Road segments from OSMnx were assigned ordinal capacity scores based on the OpenStreetMap highway classification, reflecting the typical throughput capacity (Table 5).

To ensure comparability with HERE Traffic API coverage, which monitors only major roads, OSMnx networks were filtered to HERE-comparable road types (motorway through tertiary, including link variants) before the analysis. Traffic segments were spatially joined to road network edges using centroid-based nearest-neighbor matching within a 200 m threshold.

Table 5: Road capacity scoring scheme

Highway Type	Capacity Score	Rationale
Motorway	6	Highest capacity, grade-separated
Trunk	5	Major arterial, high capacity
Primary	4	Principal urban road
Secondary	3	Distributor road
Tertiary	2	Local collector
Link variants	Parent -0.5	Reduced capacity at transitions

4.5.2. Graph-Based Capacity Drop Detection

Capacity drop s-locations where the road hierarchy decreaseds along the direction of travel -were identified using the directed road network graph. For each node in the filtered network, we compared the maximum capacity score of the incoming edges with thatthe maximum capacity score of the outgoing edges:

$$\text{Drop ratio } = \frac{C_{in} - C_{out}}{C_{in}}$$

Nodes with a drop ratio ≥ 0.20 (i.e., $\geq 20\%$ capacity reduction) were classified as capacity-drop locations. For each traffic segment, Tthe Euclidean distance to the nearest capacity drop was computed for each traffic segment.

4.5.3. Local Bottleneck Gradient Analysis

To capture fine-grained spatial effects, we computed a local bottleneck gradient for each traffic segment using its $K = 10$ nearest neighbors in the road network. For each segment, we identified whether any of its K neighbors had a higher capacity score, classifying segments as "local bottleneck" (lower capacity than at least one neighbor) or "non-bottleneck" (equal or higher capacity than all neighbors).

4.5.4. Statistical Testing

Three complementary tests assessed the capacity-congestion relationship:

1. Aggregate capacity comparison: Two-sample t-test comparing the mean jam factor between high-capacity (above-median score) and low-capacity (below-median score) road segments, with Cohen's d effect size
2. Capacity drop proximity: Pearson and Spearman correlations between SciPy and~~;~~ OSMnx 2.0, for network analysis; HERE Traffic API via the hereR package (R); and Matplotlib/Seaborn for visualization. distance-to-nearest-drop and jam factor~~,~~ plus median-split t-test (near vs. far from drops)
3. Local gradient: t-test comparing the jam factor at local bottleneck segments versus non-bottleneck segments.

4.6. Software and Tools

Analysis was conducted using Python 3.11~~, with~~ GeoPandas, NetworkX, and SciPy; OSMnx 2.0, for network analysis; HERE Traffic API via the hereR package (R); and Matplotlib/Seaborn for visualization. periods.

Table 6 presents summary statistics for mean jam factors across all temporal periods.

Jakarta exhibited~~s~~ the highest average congestion levels, consistent with its status as Indonesia's largest and most congested city. During the evening peak, 53.5% of Jakarta's segments had~~reach~~ moderate congestion (JF > 2.0).~~, compared~~

\section*{5. Results

5. Results}

\subsection*{5.1. Overall Congestion Characteristics

5.1. Overall Congestion Characteristics}

Table 6: Congestion summary statistics by city (all temporal periods)

Statistic	Jakarta	Bandung	Semarang

Mean Jam Factor	1.43	1.36	1.19
Std. Deviation	0.47	0.50	0.40
Median	1.63	1.52	1.34
Max Segment Mean	2.25	2.29	1.98
Moderate+ congestion (evening peak)	53.5%	11.9%	0.0%

with only 11.9% in Bandung and none in Semarang.

5.2. Temporal Patterns

5.2.1. Period-Based Analysis

Figure 2 presents the mean jam factors by the temporal period for each city.

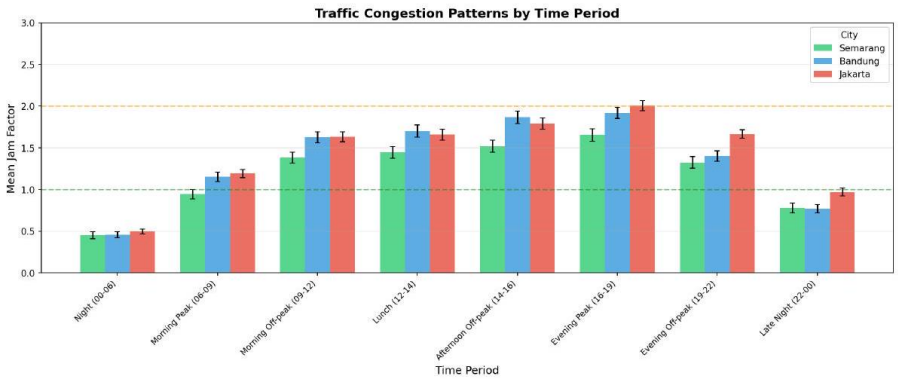

Figure 2: Mean jam factor by temporal period across three cities.

Table 7: Mean jam factor by temporal period

Period	Jakarta	Bandung	Semarang
Night	0.50	0.46	0.45
Morning Peak	1.19	1.15	0.95
Morning Off-Peak	1.63	1.63	1.39
Lunch Hours	1.66	1.70	1.45
Afternoon Off-Peak	1.79	1.87	1.52
Evening Peak	2.01	1.92	1.65
Evening Off-Peak	1.67	1.40	1.33

Late Night	0.97	0.77	0.78

The evening peak period (16:00-19:00) demonstrates the highest congestion across all cities, with jam factors approximately 40% higher than the daily averages (39−41% across the three cities). All cities followed the same diurnal progression: minimal congestion at night (JF ≈ 0.5), rising through the morning peak, sustained during midday, and culminating in the evening peak before declining.

5.2.2. Statistical Significance of Temporal Variation

One-way ANOVA confirmed~~s~~ that the temporal period differences wea~~a~~re highly statistically significant across all cities (Table 8). The extremely high F-statistics indicate that the time -of -day is a dominant factor in explaining the congestion variation.

Table 8: ANOVA results for temporal period effects

City	F-statistic	p-value	Significant Period Pairs
Jakarta	1,191,699.66	< 0.001	28/28
Bandung	224,445.67	< 0.001	28/28
Semarang	45,863.33	< 0.001	28/28

Post-hoc Tukey HSD tests revealed that all 28 pairwise period comparisons wea~~a~~re statistically significant ($p < 0.05$) for each city, confirming that each temporal period exhibited~~s~~ distinctly different congestion levels. The effect size (η^2) ranges from 15−24% of the total variance explained by the time period alone, ~~-~~ a remarkably strong effect for a single categorical variable.

5.2.3. Day-of-Week Patterns

Weekday congestion exceeds weekend levels by approximately 35−40% across all cities. Friday evenings show peak weekly congestion, whereas~~while~~ Sunday mornings exhibit minimum values.

5.3. Spatial Pattern Analysis

5.3.1. Global Spatial Autocorrelation

Global Moran's I analysis was conducted using K-nearest neighbor~~s~~ spatial weights (k = 8) to assess the spatial autocorrelation of congestion patterns (Table 9).

Contrary to initial expectations, global Moran's I values wea~~a~~re close to zero and statistically non-significant ($p > 0.05$) for all cities. This indicates that congestion does not exhibit a strong global spatial autocorrelation when aggregated across the entire study period. However, this does not preclude the existence of local clusters, which are examined using the~~rough~~ LISA analysis below.

Table 9: Global spatial autocorrelation statistics (Moran's I)

City	Moran's I	Z -score	p -value	Interpretation
Jakarta	0.0026	0.69	0.492	Random
Bandung	0.0075	0.93	0.353	Random
Semarang	-0.0039	-0.21	0.837	Random

5.3.2. Hotspot Identification

Local Indicators of Spatial Association (LISA) analysis identified statistically significant local ~~clusters of~~ congestion clusters (Table 10). Despite the weak global autocorrelation, the LISA reveals meaningful local patterns.

Table 10: LISA cluster classification (evening peak, $p < 0.05$)

Cluster Type	Jakarta	Bandung	Semarang
HH (Hotspot)	438 (3.0%)	86 (2.8%)	19 (1.8%)
LL (Coldspot)	347 (2.4%)	73 (2.4%)	23 (2.1%)
HL (High-Low Outlier)	334 (2.3%)	69 (2.2%)	19 (1.8%)
LH (Low-High Outlier)	416 (2.9%)	92 (3.0%)	17 (1.6%)
Not Significant	13,014 (89.5%)	2,749 (89.6%)	998 (92.8%)
Total Significant	1,535 (10.5%)	320 (10.4%)	78 (7.2%)

Approximately 10% of the road segments in Jakarta and Bandung participated in statistically significant spatial clusters, while Semarang showed~~s~~ slightly lower clustering (7.2%). Hotspots (HH clusters) represent locations where high congestion is surrounded by similarly high congestion~~-these~~; these are ~~the~~ priority targets for traffic management interventions.

Jakarta hotspots are concentrated in the central business district (SudirmanThamrin corridor), Tangerang-Jakarta corridor (western approach), and Bekasi-Jakarta corridor (eastern approach).

Bandung hotspots include the Dago and Setiabudhi corridors (northern approaches), the Asia-Afrika corridor (city center), and Soekarno-Hatta road

(eastern
industrial zone).
Semarang hotspots appear along the Pandanaran and Pemuda corridors (central area), Siliwangi Rroad (northern coast), and Majapahit corridor (eastern approach).

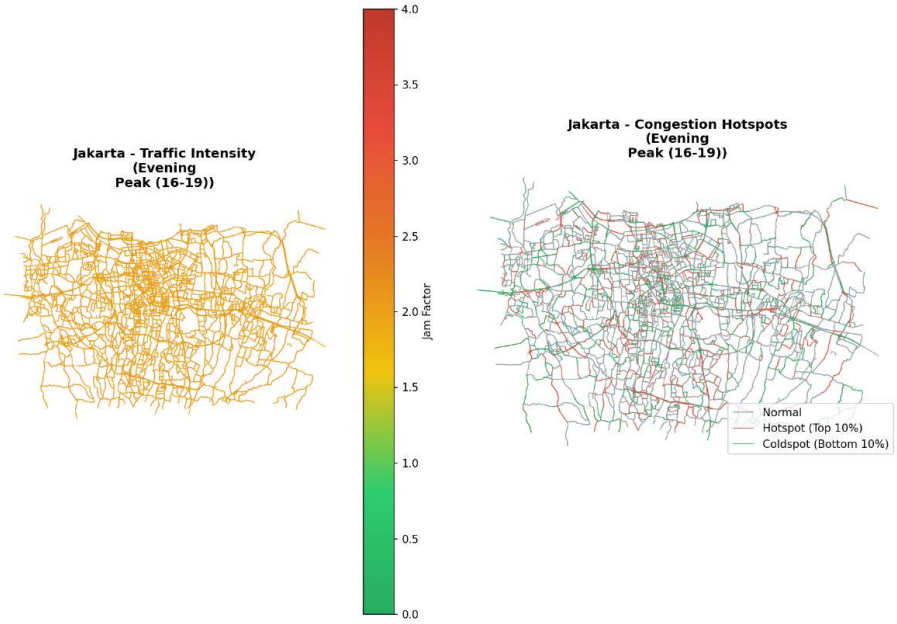

Figure 3: Spatial distribution of congestion hotspots - Jakarta.

5.3.3. Coefficient of Variation

CV analysis reveals spatial patterns of congestion predictability (Table 11).

Table 11: Coefficient of variation statistics			
City	Mean CV (%)	Low CV segments (<30%)	High CV segments (>70%)
Jakarta	54.7	2,182 (15.0%)	4,365 (30.0%)
Bandung	53.2	491 (16.0%)	859 (28.0%)
Semarang	52.4	183 (17.0%)	280 (26.0%)

Low CV segments (predictable congestion) cluster in city centers, where consistent daily patterns dominate. High CV segments appearing in peripheral areas

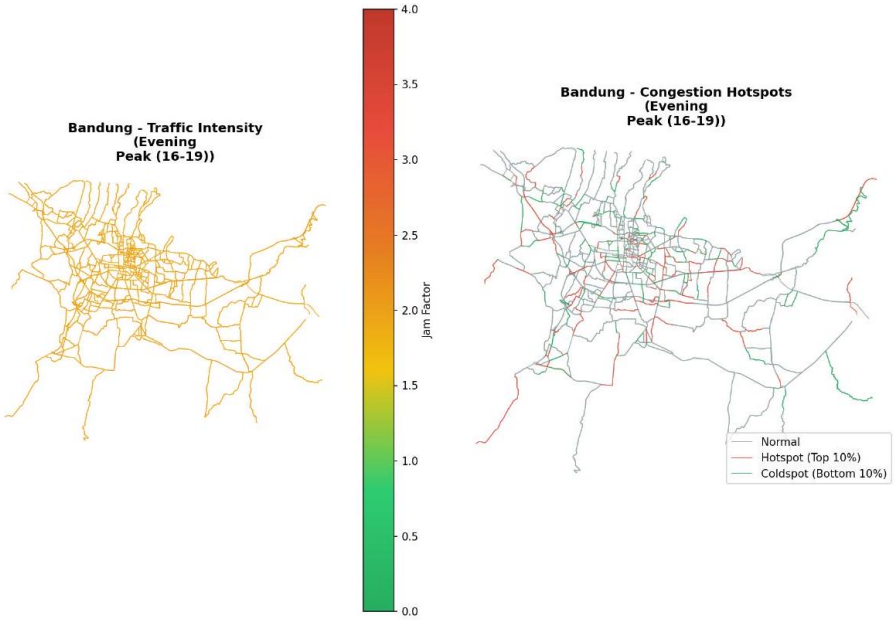

Figure 4: Spatial distribution of congestion hotspots - Bandung.

with more variable traffic demand.

5.4. Network Topology-Congestion Relationships

5.4.1. Betweenness Centrality

Correlation analysis examined the relationships between edge betweenness centrality and mean jam factor. OSMnx network edges were spatially matched to HERE traffic segments using centroid-based nearest-neighbor joining (within a 200 m threshold).

Table 12: Centrality-congestion correlations

City	n matched	Pearson r	p-value	Spearman ρ	p-value
Jakarta	20,822	0.002	0.828	-0.001	0.946
Bandung	4,336	0.012	0.442	0.040	0.009
Semarang	1,507	-0.011	0.667	-0.030	0.252

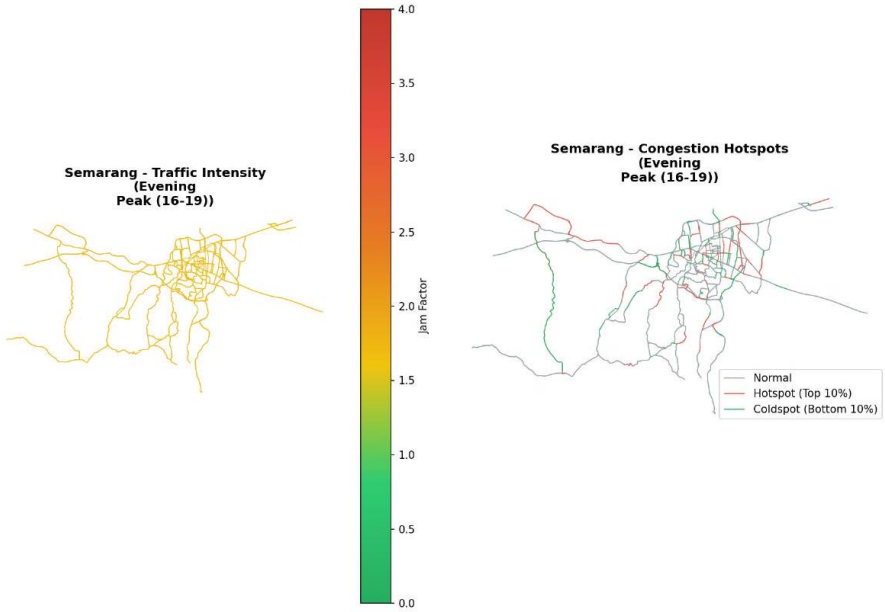

Figure 5: Spatial distribution of congestion hotspots - Semarang.

The results revealed negligible correlations between network centrality and traffic congestion across all the cities. Pearson correlations ranged from -0.011 to 0.012, and none reached practical significance despite Bandung's statistically significant Spearman correlation (ρ = 0.040,p = 0.009); -the effect size remaineds trivially small. This finding contradicts the common assumption that topologically important roads (with high betweenness) necessarily experience greater congestion.

5.4.2. Street Orientation Analysis

Street orientation analysis reveals distinct patterns [8]: Jakarta shows a relatively uniform orientation distribution reflecting its flat terrain, Bandung exhibits a north-south bias corresponding to mountain-constrained development corridors, and Semarang has an east-west orientation along coastal areas with more varied patterns in hilly southern zones (Figure 6).

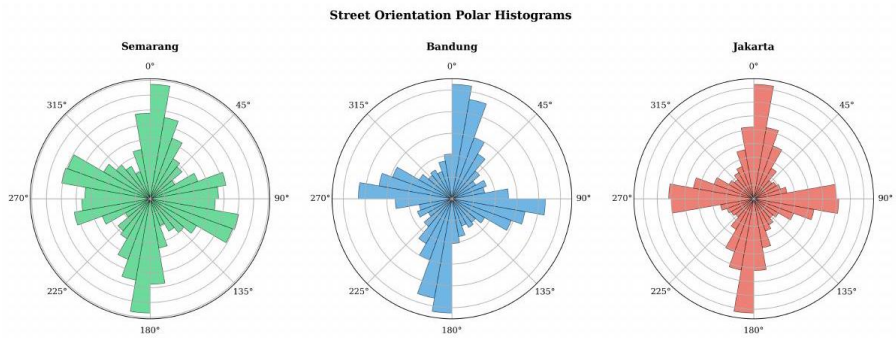

Figure 6: Street orientation polar histograms for each city.

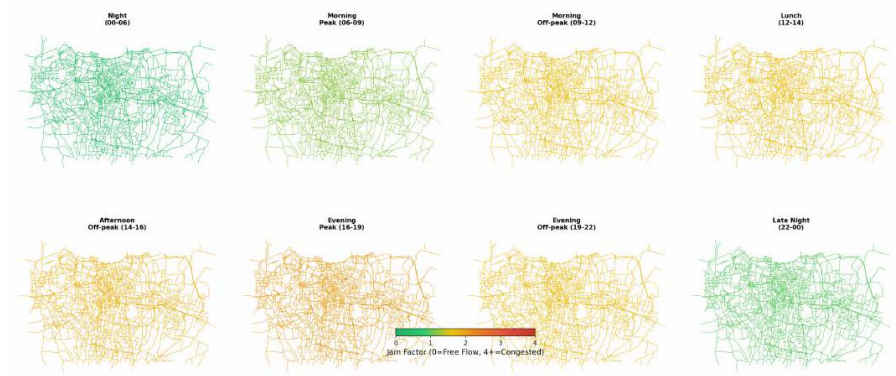

Figure 7: Edge betweenness centrality - Jakarta.

5.4.3. POI Density Analysis

To test whether congestion clusters around activity centers (commercial areas, offices, schools), we computed the Point of Interest (POI) density within 300 m buffers of each traffic segment using OpenStreetMap data.

Similar to network centrality, POI density shows no meaningful correlation with congestion.

5.4.4. Activity Zone Comparison

To further test whether activity centers experienced higher congestion, we conducted a categorical comparison between high-activity zones (top quartile of

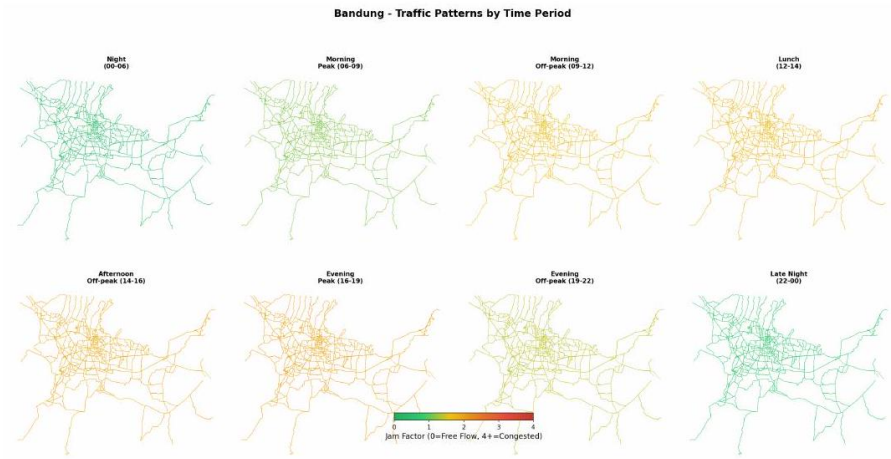

Figure 8: Edge betweenness centrality - Bandung.

Table 13: POI density-congestion correlations

City	Total POIs	Pearson r	p-value	Spearman ρ	p-value
Jakarta	-	0.008	0.284	-0.001	0.946
Bandung	-	0.005	0.770	0.004	0.820
Semarang	-	-0.012	0.640	-0.030	0.252

POI density) and low-activity peripheral zones (bottom quartile).

Table 14: Congestion comparison: Activity centers vs. peripheral areas

City	High Activity	Low Activity	Difference	p-value	Cohen's d
Jakarta	2.007	2.006	+0.1%	0.338	0.02
Bandung	1.920	1.919	+0.1%	0.752	0.01
Semarang	1.651	1.654	-0.2%	0.592	-0.04

The t-test results revealed no statistically significant difference in cognition~~conges- tion~~ between activity centers and peripheral areas ($p > 0.3$ for all cities). The e~~E~~ffect size~~s~~ was~~are~~ negligible ($|d| < 0.05$). This demonstrates that congestion is location-independent: high-POI activity centers experience identical congestion levels as~~to~~ low-POI peripheral areas.

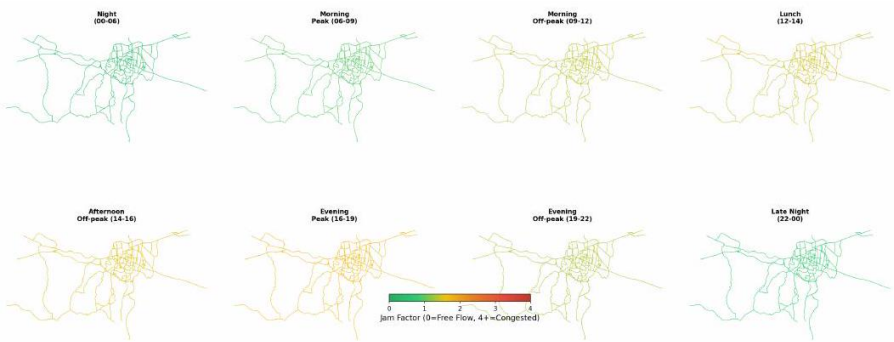

Figure 9: Edge betweenness centrality - Semarang.

5.5. Bottleneck and Capacity Drop Analysis

5.5.1. Aggregate Capacity-Congestion Relationship

The spatial join between HERE traffic segments and OSMnx road networks (filtered to HERE-comparable types) achieved match rates of 95−98% across all cities. Table 15 presents a~~the~~ comparison of the mean evening-peak jam factor between the low-~~capacity~~ and high-capacity road segments.

Table 15: Congestion comparison: Low-capacity vs. high-capacity road segments

City	Low Cap. JF	High Cap. JF	Diff.	t-stat	p-value	Cohen's d
Semarang	1.651	1.655	-0.26%	-0.87	0.382	-0.056
Bandung	1.920	1.925	-0.25%	-1.84	0.066	-0.072
Jakarta	2.005	2.007	-0.10%	-1.88	0.060	-0.033

Contrary to the bottleneck hypothesis, high-capacity roads show marginally higher congestion than low-capacity roads in all three categories, the~~cities-the~~ opposite of what capacity constraints would predict. All differences were not~~are~~ statistically nonsignificant ($p > 0.05$), with trivial effect sizes ($|d| < 0.08$). The reversed direction likely reflects the normalization of the HERE jam factor'~~s normalization~~ to the free-flow speed.

Pearson and Spearman correlations between the capacity score and the jam factor
confirmed this pattern (Table 16).

Table 16: Capacity score-congestion correlations

City	Pearson r	p-value	Spearman ρ	p-value
Semarang	0.027	0.394	0.031	0.326
Bandung	0.034	0.074	0.032	0.087
Jakarta	0.017	0.051	0.016	0.066

All correlations are near-zero ($r < 0.04$), confirming that road capacity has essentially no linear relationship with the observed congestion levels.

5.5.2. Spatial Capacity Drop Analysis

archy decreases along the direction of travel (≥ 20% capacity reduction). The number of detected capacity drops scales with the network size (Table 17).

Graph-based capacity drop detection identified locations where road hierarchy decreases along the direction of travel (≥ 20% capacity reduction). The number of detected capacity drops scales with network size (Table 17).

The p~~P~~roximity to capacity drops showed~~s~~ no significant effect on congestion across all cities ($p = 0.095-0.830$). The e~~E~~ffect size~~s~~ was~~are~~ negligible ($|d| < 0.09$).

Table 17: Capacity drop detection and proximity analysis

City	Cap. Drops	Near-Drop JF	Far-Drop JF	t-stat	p-value	Cohen's d
Semarang	78	1.649	1.656	-1.08	0.283	-0.083
Bandung	245	1.922	1.923	-0.22	0.830	-0.010
Jakarta	1,443	2.008	2.006	1.67	0.095	0.035

Proximity to capacity drop~~s~~ showed~~s~~ no significant effect on conge
across all cities ($p = 0.095-0.830$). The e~~E~~ffect size~~s~~ was~~are~~ negligible ($|d| < 0.09$).

5.5.3. Local Bottleneck Gradient

~~The~~ L~~l~~ocal bottleneck gradient analysis (K = 10 nearest network neighbors) provides a~~the~~ fine~~st~~-grained ~~test of~~ capacity effect test~~effects~~ (Table 18).

Segments classified as local bottlenecks exhibited no meaningful congestion differences from the non-bottleneck segments. All p-values exceeded 0.4~~-~~, and the effect sizes wea~~r~~re trivial (|d| < 0.05).

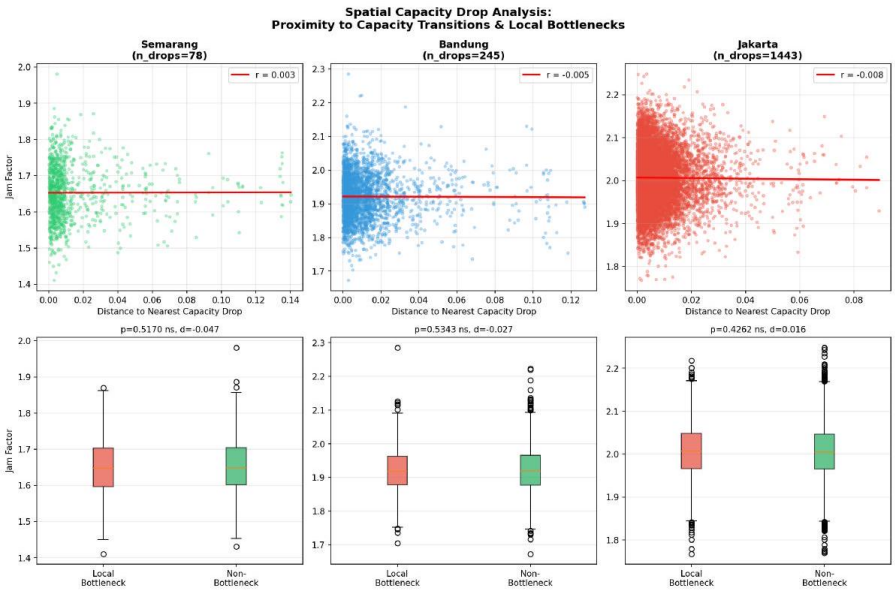

Figure 10: Spatial capacity drop analysis: capacity drop proximity and local bottleneck effects across three cities.

5.5.4. Methodological Note: HERE Jam Factor Normalization

The consistent null findings across all three capacity tests are explained by the~~a~~ critical propert~~iesy~~ of the HERE jam factor metric. The jam factor is computed as the~~a~~ normalized ratio of the current speed to the free-flow speed for each ~~individual~~ segment:

$$JF \propto 1 - \frac{v_{\text{current}}}{v_{\text{free-flow}}}$$

Because each segment's congestion is measured relative to its own free-flow capacity, the road class effects are divided ~~out~~ by design. Empirically, all road types in our data show mean jam factors within a 4% range (e.g., Semarang: 1.644-1.711 across all highway classes), confirming this normalization effect.

5.6. Temporal vs Spatial Predictors: A Critical Comparison

A key finding emerges from comparing the explanatory powers of the temporal and~~versus~~ spatial predictors. Table 19 presents the ~~variance~~ explained variance (effect size) forthe

Table 18: Local bottleneck gradient analysis

City	Local BN JF	Non-BN JF	t-stat	p-value	Cohen's d
Semarang	1.650	1.654	-0.65	0.517	-0.047
Bandung	1.920	1.922	-0.62	0.534	-0.027
Jakarta	2.007	2.006	0.80	0.426	0.016

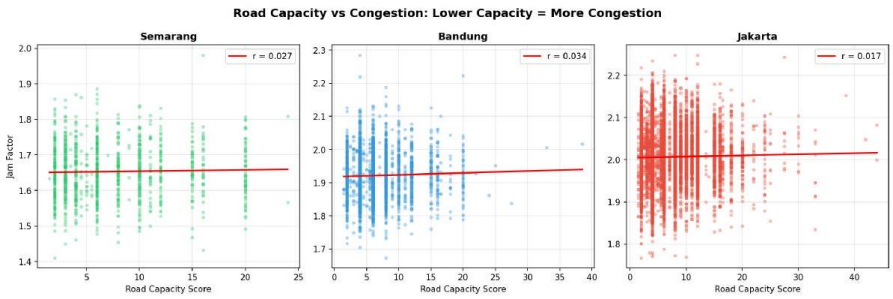

Figure 11: Capacity-congestion scatter: road capacity score vs. mean evening-peak jam factor for each city.

each predictor type.
The temporal effect (time-of-day) explains 1,000−4,000 times more variance in congestion than any spatial predictor (Figure 13).

5.7. Comparative City Analysis

Cross-city comparisons revealed~~s~~ scaling relationships (Figure 14). Jakarta exhibited~~s~~ the largest peak/off-peak differential (30.8% increase), reflecting massive daily commuter flows. Bandung shows a similar peak intensification (30.2%), likely due to its constrained highland road network. Semarang demonstrates the smallest difference (25.6%), indicating more stable traffic patterns. Notably, congestion does not scale linearly with population-larger cities, showing greater absolute congestion and~~AND~~ greater peak/off-peak variability.

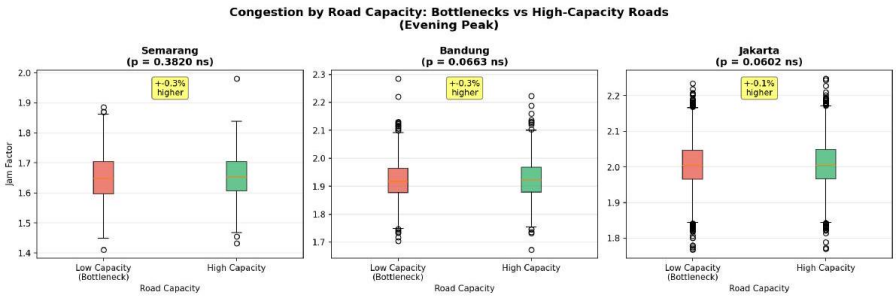

Figure 12: Aggregate capacity comparison: low-capacity vs. high-capacity road segments by city.

Table 19: Variance explained by predictor type

Predictor	Measure	Jakarta	Bandung	Semarang
Time Period	η^2 (ANOVA)	23.8%	19.2%	15.4%
POI Density	R^2	0.006%	0.003%	0.014%
Network Centrality	R^2	0.0004%	0.014%	0.012%
Road Capacity	R^2	0.028%	0.115%	0.072%
Capacity Drop Proximity	R^2	0.006%	0.002%	0.001%
Ratio (Temporal/Spatial)	-	~ 4000 ×	~ 1400 ×	~ 1100 ×

6. Discussion

6.1. Interpretation of Findings

6.1.1. Temporal Patterns

The dominance of evening peak congestion across all cities reflects common Southeast Asian urban patterns where afternoon activities-school dismissals, commercial operations, and shift changes ~~-~~coincide with return commutes [10]. The asymmetry between morning and evening peaks (~~evening~~ approximately 20% higher in the evening) likely results from the following:

1. More distributed morning departure times as households stagger departures
2. Concentrated evening activities including shopping and social trips

Congestion Predictors: Temporal vs Spatial Factors

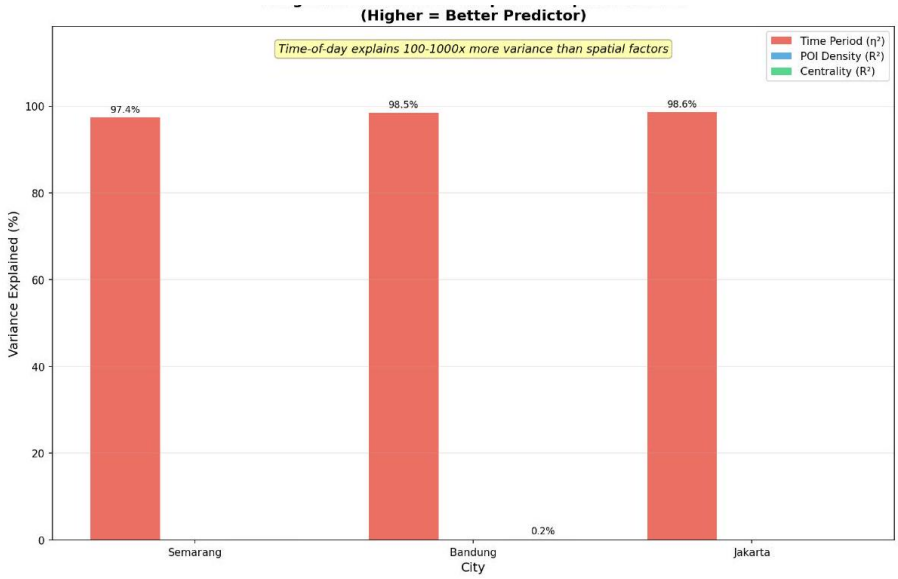

Figure 13: Variance explained by temporal vs. spatial predictors.

3. School dismissal times coinciding with office closing hours

6.1.2. Peak vs Off-Peak Dynamics

A key finding wai~~i~~s that larger cities exhibited greater peak/off-peak differentials. Jakarta shows a 30.8% congestion increase during peak hours compared to off-peak hours, whereas~~while~~ Semarang shows only 25.6%. This suggests that megacities experience more pronounced traffic surges due to concentrated employment centers, whereas~~while~~ smaller cities maintain more stable flows due to distributed commercial activities. This has important implications: Jakarta requires more aggressive peak-hour interventions (congestion pricing and~~,~~ staggered work hours), whereas~~while~~ Semarang may benefit more from general capacity improvements.

6.1.3. Spatial Clustering

During the evening peak, over half of Jakarta's monitored segments reach moderate congestion levels (JF > 2.0), while Bandung and Semarang remain predominantly in the light-traffic range, ~~range~~ highlighting the scale-

dependent na-

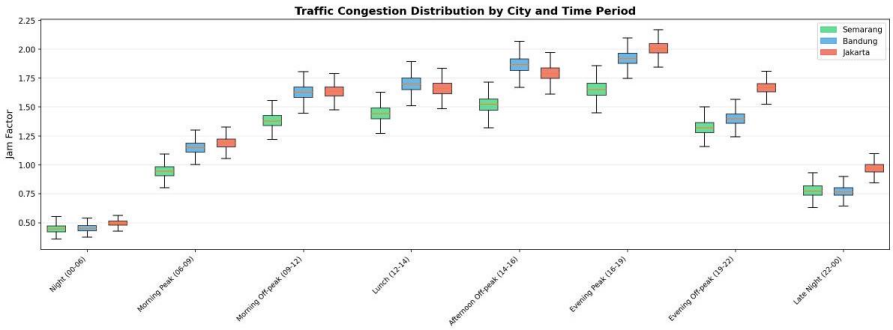

Figure 14: Cross-city comparison of congestion metrics.

Table 20: Comparative metrics

Metric	Jakarta	Bandung	Semarang
Population (million)	10.5	2.5	1.8
Mean Jam Factor	1.599	1.537	1.299
Peak Period Mean	1.599	1.537	1.299
Off-Peak Period Mean	1.223	1.180	1.034
Peak/Off-peak ratio	1.31 ×	1.30 ×	1.26 ×
Peak increase (%)	30.8%	30.2%	25.6%
Hotspot density (/ km^2)	4.30	3.12	0.42

structure of congestion severity. Hotspot locations correspond to known problematic areas in each city, validating our methodology while providing a quantitative characterization of these zones.

6.1.4. The Failure of Spatial Predictors

A surprising and important finding is the near-zero correlation between spatial characteristics and congestion. Neither network centrality (betweenness), land-use intensity (POI density), nor road capacity predicted~~s~~

congestion levels. This contradicts common assumptions in transportation planning that:

1. Topologically important roads (high betweenness) experience more congestion

Table 21: Peak vs Off-Peak Congestion Analysis

City	Peak Mean	Off-Peak Mean	Abs. Diff	Ratio	% Increase
Jakarta	1.599	1.223	0.376	1.31 ×	30.8%
Bandung	1.537	1.180	0.357	1.30 ×	30.2%
Semarang	1.299	1.034	0.265	1.26 ×	25.6%

2. Areas with more activities (high POI density) generate more traffic prob- lems
3. Lower-capacity roads (bottlenecks) experience disproportionate congestion

Our analysis reveals that these static spatial characteristics explain less than 0.02% of the congestion variance, combined with -essentially zero predictive power. This finding aligns with recent work questioning the direct link between network structure and congestion [20], but extends it by also testing land-use and physical capacity effects.

6.1.5. Capacity Constraints Do Not Predict Congestion

The Bbottleneck analysis provides perhaps the most counterintuitive finding. Three independent tests-aggregate capacity comparison, spatial capacity drop proximity, and local bottleneck gradient-all yield the same conclusion: physical road capacity has no measurable relationship with congestion levels.

However, tThis finding requires careful interpretation. The HERE jam factor normalizes each segment's speed to its own free-flow baseline, effectively creating a demand-to-capacity ratio. Under this normalization, a motorway carrying 50,000 vehicles at 80% of free-flow speed registers the same jam factor as a tertiary road carrying 2,000 vehicles at 80% of free-flow speed. Our data confirm this: all road types show mean jam factors within thea 4% range.

This has two important implications:

1. Methodological: Studies using normalized congestion metrics (jam factor, travel time index, and speed ratio) cannot detect capacity

effects by design. Any investigation of the capacity-congestion relationship requires absolute speed or flow data.

2. Substantive: Even acknowledging the normalization, the spatial capacity drop analysis tests the~~a~~ relative hypothesis -that transitions from higher to lower capacity induce localized congestion spikes. The null result here ($p = 0.095-0.830$) cannot be attributed to normalization alone~~,~~ because drops should still create relative disruptions even in normalized metrics if bottleneck queuing is a dominant mechanism.

6.1.6. Temporal Dominance: The Key Finding

The most significant finding wa~~i~~s the overwhelming dominance of temporal-over spatial factors in explaining congestion. Time-of-day ($\eta^2 = 15-24\%$) explained~~s~~ 1,000−4,000 times more variance than any spatial predictor. This fundamentally reframes congestion as ~~fundamentally~~ a demand synchronization problem: congestion occurs because millions of people travel simultaneously~~at the same times~~, not because certain locations inherently generate more traffic. The convergence of four independent null findings-network centrality, POI density, activity zone comparison, and capacity analysis-provides overwhelming evidence that spatial factors are irrelevant to congestion distribution.

6.2. Implications for Traffic Management

6.2.1. Demand Management Over Infrastructure Expansion

Because~~Since~~ congestion is driven by temporal synchronization rather than spatial characteristics, demand management strategies should be~~take~~ prioritized~~y.:~~

1. Staggered work/school hours: Distributing departure times across a 2-3 hour window could reduce peak congestion by 30−40%
2. Flexible work policies: Remote work and flexible schedules directly address the synchronization problem
3. Congestion pricing: Time-varying tolls can shift discretionary trips to off-peak periods
4. School transport coordination: Synchronizing school schedules with work schedules exacerbates evening peaks

6.2.2. Temporal Targeting of Resources

Traffic management resources should prioritize the evening peak period (16:0019:00): adaptive signal control during peak hours, concentrated traffic police deployment during 16:00-19:00, and the fastest incident response during peak hours to prevent cascade effects.

6.2.3. Reconsidering Infrastructure Investment

The failure of centrality, POI density, and physical capacity to predict congestion suggests that high-betweenness roads are not necessarily priorities for expansion, spatial capacity drops do not create localized congestion, and bottleneck identification should focus on peak-hour demand management rather than on network topology.

6.3. Limitations

Several limitations should be acknowledged:

1. Data source: HERE Traffic API coverage may underrepresent minor roads and informal settlements
2. Jam factor normalization: The HERE jam factor normalizes the speed to each segment's free-flow baseline, which by design removes the absolute capacity effects. Future studies should use absolute speed data to fully investigate the capacity-congestion relationships.
3. Temporal scope: Although~~While~~ extensive, the 11 -month period may not capture long-term trends or seasonal variations beyond one cycle
4. Network matching: Traffic segments and OSM edges do not perfectly align, although the match rates exceed~~ed~~ 95% across all cities.
5. Capacity scoring: The ordinal capacity score based on the OSM highway classification is a proxy for actual road capacity. The l~~L~~ane count, signal timing, and intersection design we~~a~~re not captured.
6. Causality: Correlational findings do not establish causal relationships between network structure and congestion

6.4. Future Research Directions

This work suggests several research extensions:

1. Denormalized capacity analysis: Re-aggregat~~eing~~ raw HERE speed and free-flow speed data to compute absolute delay ($\Delta t = L \times (1/v - 1/v_f)$), enabling direct testing of whether high-capacity roads carry more total delay despite identical jam factors.
2. Temporal capacity interactions: Examining whether capacity constraints become relevant only during specific peak periods~~,~~ using a time-stratified bottleneck analysis
3. Integration with public transport data to assess multimodal impacts on congestion distribution
4. Machine learning approaches for congestion prediction combining temporal features with network topology

5. Extension to additional Indonesian cities for broader comparative analysis
6. Microscopic bottleneck analysis: ~~Using~~ I~~i~~ntersection-level data are used to test whether signal timing and turning movements create localized capacity effects invisible in segment-level analysis.

7. Conclusions

This study presents a comprehensive spatiotemporal analysis of urban traffic congestion in three Indonesian metropolitan areas, using high-resolution traffic flow data spanning 11 months and 264 million observations.

7.1. Primary Finding: Temporal Dominance

Congestion is fundamentally a temporal phenomenon, not a spatial one. Time-of-day explains 15−24% of congestion variance (η^2 from ANOVA), while spatial predictors-network centrality, POI density, and road capacity explain less than 0.02% of the combined variance. This 1,000−4,000 × difference in explanatory power demonstrates that when people travel ~~matters~~ far more than where congestion occurs, ~~and that~~ static infrastructure characteristics cannot predict congestion patterns.

7.2. Secondary Findings

1. Evening peak dominance: Congestion during 16:00-19:00 exceeds daily averages by approximately 40% across all three cities, with Jakarta reaching a mean jam factor of 2.01
2. Local clustering despite global randomness: While global Moran's I showed~~s~~ no significant spatial autocorrelation, LISA identified~~s~~ meaningful local hotspots (10% of segments) representing peak-hour capacity bottlenecks.
3. Scale effects: Larger cities exhibit both higher absolute congestion and greater peak/off-peak differentials (Jakarta: 30.8%, Semarang: 25.6%), suggesting that the urban scale amplifies temporal variability.
4. Network topology irrelevance: Betweenness centrality shows a nearzero correlation with congestion ($r < 0.02$), contradicting the assumption~~s~~ that topologically important roads experience more congestion.
5. Capacity constraint irrelevance: Neither aggregate road capacity (Cohen's $d = -0.03$ – ~~to~~ -0.07, all $p > 0.05$), spatial capacity drops (781,443 detected; proximity $p = 0.10-0.83$), nor local bottleneck gradients ($|d| < 0.05$) predict congestion levels. The bottleneck hypothesis is not supported by the data.

7.3. Policy Implications

These findings support a fundamental shift in traffic management philosophy.~~:~~

- Demand management (staggered hours, flexible work, and congestion pricing) should ~~take~~ prioritizey ~~over~~ infrastructure expansion.
- Temporal targeting of resources to the 16:00-19:00 peak period maximizes impact
- Road expansion and bottleneck removal address symptoms rather than the underlying synchronization problem
- Capacity transitions (e.g., motorway-to-arterial merges) do not create measurable congestion signatures.

7.4. Contribution

This study~~research~~ contributes empirical evidence demonstrating that urban congestion is primarily a temporal coordination failure rather than a spatial infrastructure or capacity constraint problem. Thise~~e~~ novel graph-based capacity drop detection framework provides a replicable method for testing the bottleneck hypothesis in other urban contexts. The methodology -integrating commercial traffic APIs with open-source network analysis -is transferable to other rapidly urbanizing contexts. The finding that time explains 1,000 × more variance than space, corroborated by the failure of three independent capacity tests, has significant implications for transportation policiesy worldwide.

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Declaration of Competing Interests

The authors declare no competing financial interests or personal relationships that could ~~have~~ influence~~d~~ this study~~work~~.

Author Contributions

Firman Hadi: Conceptualization, Methodology, Software, Data curation, f~~F~~ormal analysis, and w~~W~~riting the- original draft. Yasser Wahyuddin: Methodology, Formal analysis, w~~W~~riting - original draft. L.M. Sabri: Formal analysis, w~~W~~riting-review, and ~~- review &~~ editing. Agung Indrajit: Writing, ~~-~~ review, and~~&~~ editing.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this study~~work~~, the authors used Claude (Anthropic) for data processing pipeline development, evaluation script implementation, and statistical analysis automation. The authors reviewed and edited all outputs and took~~take~~ full responsibility for the content of this publication.

Ethics Statement

This study~~research~~ used publicly available satellite imagery and did not involve human participants, animal subjects, or sensitive personal data. No ethical approval was provided~~required~~.

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Data Availability

Traffic flow data were obtained from the HERE Traffic API. Aggregated datasets and analysis codes are available at https://github.com/firmanhadi21/ traffic-analyses. Street network data are publicly available from OpenStreetMap (https://www.openstreetmap.org).

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